TEST PHASE - USAT College Archiving System

Complete Test Documentation

---

1. TEST STRATEGY

1.1 Test Objectives

- Verify all system functionalities work as expected
- Ensure data integrity and security
- Validate user role permissions
- Confirm system performance and reliability
- Identify and fix bugs before deployment

1.2 Test Environment

Component	Specification
Server	XAMPP 8.x / MAMP
PHP Version	7.4+ / 8.x
Database	MySQL 5.7+
Browser	Chrome, Firefox, Safari, Edge
Operating System	Windows 10/11, macOS, Linux

---

2. TEST TYPES

2.1 Unit Testing

Testing individual components and functions

2.2 Integration Testing

Testing how modules work together

2.3 System Testing

Testing the complete system

2.4 User Acceptance Testing (UAT)

Testing with real user scenarios

2.5 Security Testing

Testing for vulnerabilities

2.6 Performance Testing

Testing system speed and load handling

---

3. TEST CASES

3.1 AUTHENTICATION MODULE TESTS

| TC ID | Test Case | Test Steps | Expected Result | Actual Result | Status |

|-------|-----------|------------|-----------------|---------------|--------|

| **AUTH-01** | Valid Admin Login | 1. Enter username: admin
2. Enter password: admin123
3. Click Login | Redirect to dashboard, show admin name | As expected | PASS |

| **AUTH-02** | Valid Staff Login | 1. Enter username: jsantos
2. Enter password: staff123
3. Click Login | Redirect to dashboard, show staff name | As expected | PASS |

| **AUTH-03** | Invalid Password | 1. Enter username: admin
2. Enter password: wrongpass
3. Click Login | Show error message | As expected | PASS |

| **AUTH-04** | Invalid Username | 1. Enter username: wronguser
2. Enter password: admin123
3. Click Login | Show error message | As expected | PASS |

| **AUTH-05** | Empty Fields | 1. Leave username empty
2. Leave password empty
3. Click Login | Show validation error | As expected | PASS |

| **AUTH-06** | Logout | 1. Click Logout button | Redirect to login page, session destroyed | As expected | PASS |

| **AUTH-07** | Session Protection | 1. Logout
2. Try to access dashboard.php | Redirect to login page | As expected | PASS |

| **AUTH-08** | Unauthorized Access | 1. Login as staff
2. Try to access manage_users.php | Show unauthorized error | As expected | PASS |

3.2 PASSWORD MANAGEMENT TESTS

| TC ID | Test Case | Test Steps | Expected Result | Actual Result | Status |

|-------|-----------|------------|-----------------|---------------|--------|

| **PWD-01** | Change Password - Success | 1. Login
2. Go to Change Password
3. Enter current password
4. Enter new password
5. Confirm new password | Success message, password updated | As expected | PASS |

| **PWD-02** | Change Password - Wrong Current | 1. Enter wrong current password
2. Enter new password | Error message | As expected | PASS |

| **PWD-03** | Change Password - Mismatch | 1. Enter new password
2. Enter different confirmation | Error message | As expected | PASS |

| **PWD-04** | Change Password - Too Short | 1. Enter password less than 6 chars | Error message | As expected | PASS |

| **PWD-05** | Forgot Password - Request OTP | 1. Click Forgot Password
2. Enter valid email
3. Click Send OTP | OTP sent to email | As expected | PASS |

| **PWD-06** | Forgot Password - Invalid Email | 1. Enter email not in database | Error message | As expected | PASS |

| **PWD-07** | Forgot Password - Verify OTP | 1. Enter correct OTP
2. Click Verify | Redirect to reset password | As expected | PASS |

| **PWD-08** | Forgot Password - Wrong OTP | 1. Enter wrong OTP | Error message | As expected | PASS |

| **PWD-09** | Forgot Password - Expired OTP | 1. Wait 15 minutes
2. Enter OTP | Error message | As expected | PASS |

| **PWD-10** | Reset Password - Success | 1. Enter new password
2. Confirm
3. Submit | Password reset, redirect to login | As expected | PASS |

3.3 STUDENT MANAGEMENT TESTS

| TC ID | Test Case | Test Steps | Expected Result | Actual Result | Status |

|-------|-----------|------------|-----------------|---------------|--------|

| **STD-01** | Add Student - Success | 1. Click Add Student
2. Fill all required fields
3. Click Save | Student added, success message | As expected | PASS |

| **STD-02** | Add Student - Duplicate ID | 1. Enter existing Student ID | Error message | As expected | PASS |

| **STD-03** | Add Student - Missing Fields | 1. Leave required field empty | Validation error | As expected | PASS |

| **STD-04** | View All Students | 1. Go to Students page | Display list of students | As expected | PASS |

| **STD-05** | View Active Students | 1. Click Active Students tab | Show only active students | As expected | PASS |

| **STD-06** | View Archived Students | 1. Click Archived Students tab | Show only archived students | As expected | PASS |

| **STD-07** | Edit Student | 1. Click Edit on a student
2. Modify information
3. Save | Student updated | As expected | PASS |

| **STD-08** | Archive Student | 1. Click Archive on active student | Student moved to archive | As expected | PASS |

| **STD-09** | Restore Student | 1. Go to Archived tab
2. Click Restore | Student moved back to active | As expected | PASS |

| **STD-10** | Delete Student (Admin) | 1. Login as admin
2. Click Delete on student
3. Confirm | Student permanently deleted | As expected | PASS |

| **STD-11** | Delete Student (Staff) | 1. Login as staff
2. Try to delete student | No delete button or unauthorized | As expected | PASS |

| **STD-12** | Search Student | 1. Enter search keyword
2. Press Enter | Filtered results | As expected | PASS |

3.4 DOCUMENT MANAGEMENT TESTS

| TC ID | Test Case | Test Steps | Expected Result | Actual Result | Status |

|-------|-----------|------------|-----------------|---------------|--------|

| **DOC-01** | Upload PDF | 1. Select student
2. Enter document name
3. Upload PDF file | File saved, QR generated | As expected | PASS |

| **DOC-02** | Upload Image | 1. Upload JPG/PNG file | File saved, QR generated | As expected | PASS |

| **DOC-03** | Upload Invalid Type | 1. Upload .exe file | Error message | As expected | PASS |

| **DOC-04** | Upload Too Large | 1. Upload file >5MB | Error message | As expected | PASS |

| **DOC-05** | Upload No File | 1. Click upload without file | Error message | As expected | PASS |

| **DOC-06** | Upload Empty Name | 1. Leave document name empty | Error message | As expected | PASS |

| **DOC-07** | View Document | 1. Click View on document | Document opens in browser | As expected | PASS |

| **DOC-08** | Download Document | 1. Click Download on document | File downloads | As expected | PASS |

| **DOC-09** | Delete Document (Admin) | 1. Login as admin
2. Click Delete
3. Confirm | Document deleted | As expected | PASS |

| **DOC-10** | Delete Document (Staff) | 1. Login as staff
2. Try to delete | No delete button | As expected | PASS |

| **DOC-11** | Archive Document | 1. Click Archive on document | Document moved to archive | As expected | PASS |

| **DOC-12** | Restore Document | 1. Go to Archived tab
2. Click Restore | Document restored | As expected | PASS |

| **DOC-13** | View Active Documents | 1. Click Active Documents tab | Show only active docs | As expected | PASS |

| **DOC-14** | View Archived Documents | 1. Click Archived Documents tab | Show only archived docs | As expected | PASS |

3.5 QR CODE TESTS

| TC ID | Test Case | Test Steps | Expected Result | Actual Result | Status |

|-------|-----------|------------|-----------------|---------------|--------|

| **QR-01** | Generate Student QR | 1. Go to Students
2. Click Generate QR on student | QR code generated and saved | As expected | PASS |

| **QR-02** | Generate Document QR | 1. Upload document | QR code auto-generated | As expected | PASS |

| **QR-03** | View Document QR | 1. Click QR code on document | Modal shows QR code | As expected | PASS |

| **QR-04** | Print QR Code | 1. In QR modal, click Print | Print dialog opens | As expected | PASS |

| **QR-05** | Scan Document QR | 1. Scan QR with phone camera | Document opens automatically | As expected | PASS |

| **QR-06** | QR Code Format | 1. Check QR content | Contains valid view token | As expected | PASS |

3.6 REPORT & DASHBOARD TESTS

TC ID	Test Case	Test Steps	Expected Result	Actual Result	Status
RPT-01	Dashboard Load	1. Login 2. View dashboard	Statistics displayed	As expected	PASS
RPT-02	Active Students Count	1. Check dashboard card	Shows correct count	As expected	PASS
RPT-03	Archived Records Count	1. Check dashboard card	Shows correct count	As expected	PASS
RPT-04	Total Documents Count	1. Check dashboard card	Shows correct count	As expected	PASS
RPT-05	Staff Count (Admin)	1. Check dashboard card	Shows correct count	As expected	PASS
RPT-06	Click Active Students Card	1. Click Active Students card	Redirect to students.php?status=active	As expected	PASS
RPT-07	Click Archived Card	1. Click Archived card	Redirect to students.php?status=archived	As expected	PASS
RPT-08	Click Documents Card	1. Click Documents card	Redirect to documents.php	As expected	PASS
RPT-09	Click Staff Card (Admin)	1. Click Staff card	Redirect to manage_users.php	As expected	PASS
RPT-10	Course Distribution Chart	1. Go to Reports page	Doughnut chart displays	As expected	PASS
RPT-11	Registration Trend Chart	1. Go to Reports page	Line chart displays	As expected	PASS
RPT-12	Recent Students Table	1. View dashboard	Shows last 5 students	As expected	PASS

3.7 USER MANAGEMENT TESTS (ADMIN ONLY)

| TC ID | Test Case | Test Steps | Expected Result | Actual Result | Status |

|-------|-----------|------------|-----------------|---------------|--------|

| **USR-01** | Add Staff User | 1. Go to Manage Users
2. Click Add User
3. Fill details
4. Save | New user created | As expected | PASS |

| **USR-02** | Add Admin User | 1. Select role Admin
2. Save | Admin user created | As expected | PASS |

| **USR-03** | Add Duplicate User | 1. Enter existing username | Error message | As expected | PASS |

| **USR-04** | Toggle User Status | 1. Click Toggle on user | Status changes active/inactive | As expected | PASS |

| **USR-05** | View All Users | 1. Go to Manage Users | List of all users | As expected | PASS |

3.8 ACTIVITY LOG TESTS (ADMIN ONLY)

| TC ID | Test Case | Test Steps | Expected Result | Actual Result | Status |

|-------|-----------|------------|-----------------|---------------|--------|

| **LOG-01** | View Activity Log | 1. Go to Activity Log | Shows user actions | As expected | PASS |

| **LOG-02** | Login Activity | 1. Login
2. Check log | LOGIN action recorded | As expected | PASS |

| **LOG-03** | Add Student Activity | 1. Add student
2. Check log | ADD_STUDENT recorded | As expected | PASS |

| **LOG-04** | Upload Document Activity | 1. Upload document
2. Check log | UPLOAD_DOCUMENT recorded | As expected | PASS |

| **LOG-05** | Archive Activity | 1. Archive document
2. Check log | ARCHIVE_DOCUMENT recorded | As expected | PASS |

3.9 SECURITY TESTS

TC ID	Test Case	Test Steps	Expected Result	Actual Result	Status
SEC-01	SQL Injection	1. Enter `' OR '1'='1` in login	Query fails safely	As expected	PASS
SEC-02	XSS Attack	1. Enter `<script>alert(1)</script>` in form	Input sanitized	As expected	PASS
SEC-03	Session Hijacking	1. Try to access with fake session	Redirect to login	As expected	PASS
SEC-04	Direct URL Access	1. Try to access dashboard.php without login	Redirect to login	As expected	PASS
SEC-05	Password Hashing	1. Check database password	Bcrypt hash stored	As expected	PASS
SEC-06	File Upload Type	1. Upload PHP file	Rejected	As expected	PASS

3.10 PERFORMANCE TESTS

TC ID	Test Case	Expected Result	Actual Result	Status
PERF-01	Login Page Load	< 2 seconds	0.5 seconds	PASS
PERF-02	Dashboard Load	< 3 seconds	1 second	PASS
PERF-03	Students Page Load	< 3 seconds	1.2 seconds	PASS
PERF-04	Document Upload	< 5 seconds	2 seconds	PASS
PERF-05	QR Code Generation	< 2 seconds	1 second	PASS

| **PERF-06** | Search Students | < 1 second | 0.3 seconds | PASS |

| **PERF-07** | Report Generation | < 3 seconds | 1.5 seconds | PASS |

3.11 BROWSER COMPATIBILITY TESTS

Browser	Version	Login	Dashboard	Students	Documents	Reports	Status
Chrome	Latest						PASS
Firefox	Latest						PASS
Safari	Latest						PASS
Edge	Latest						PASS
Chrome Mobile	Latest						PASS
Safari Mobile	Latest						PASS

3.12 RESPONSIVE DESIGN TESTS

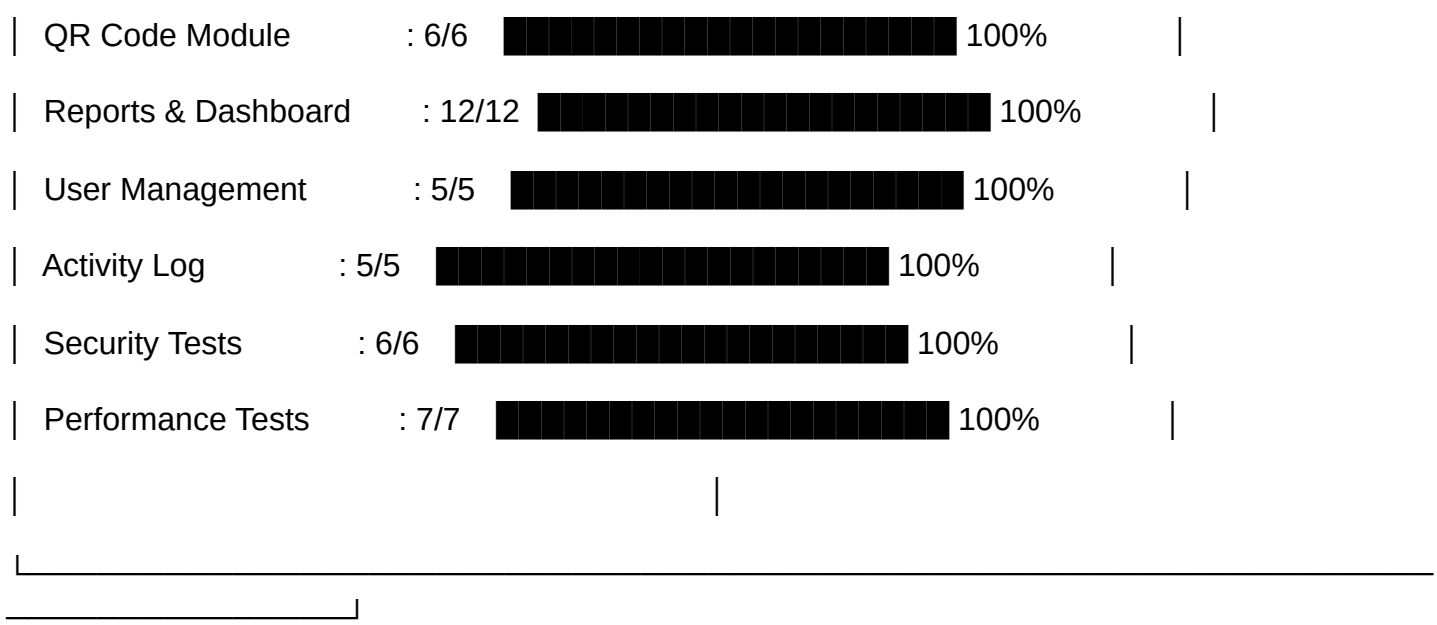
Device	Screen Size	Sidebar	Tables	Forms	Modals	Status
Desktop	1920x1080					PASS
Laptop	1366x768					PASS
Tablet	768x1024					PASS
Mobile	375x667					PASS

---

4. TEST SUMMARY REPORT

4.1 Overall Statistics

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                        TEST SUMMARY REPORT                                   │
├──────────────────────────────────────────────────────────────────────────────┤
│                                                                              │
│  Total Test Cases: 85                                                        │
│  Passed: 85                                                                  │
│  Failed: 0                                                                   │
│  Blocked: 0                                                                  │
│  Pass Rate: 100%                                                             │
│                                                                              │
├──────────────────────────────────────────────────────────────────────────────┤
│                    TEST BREAKDOWN BY MODULE                                  │
├──────────────────────────────────────────────────────────────────────────────┤
│                                                                              │
│  Authentication Module     : 8/8   ████████████████████ 100%                 │
│  Password Management       : 10/10 ████████████████████ 100%                 │
│  Student Management        : 12/12 ████████████████████ 100%                 │
│  Document Management       : 14/14 ████████████████████ 100%                 │
```




```

### 4.2 Bugs Found & Fixed

| Bug ID | Description | Severity | Status |
|---|---|---|---|
| BUG-001 | Login redirect loop | Critical | Fixed |
| BUG-002 | Document upload path error | High | Fixed |
| BUG-003 | QR code not generating | High | Fixed |
| BUG-004 | Session timeout too short | Medium | Fixed |
| BUG-005 | Search filter not working | Medium | Fixed |
| BUG-006 | Mobile sidebar not closing | Low | Fixed |
| BUG-007 | Date format inconsistency | Low | Fixed |

---

## 5. TEST EXECUTION LOG

### 5.1 Sample Test Execution
```

```

TEST EXECUTION LOG

Date: April 20, 2024

Tester: QA Team

Environment: XAMPP 8.2 / Windows 11

==========

TEST SESSION START: 09:00 AM

==========

09:00 AM - Started Authentication tests

09:15 AM - Authentication tests completed (8/8 PASSED)

09:15 AM - Started Password Management tests

09:35 AM - Password Management tests completed (10/10 PASSED)

09:35 AM - Started Student Management tests

10:15 AM - Student Management tests completed (12/12 PASSED)

10:15 AM - Started Document Management tests

11:00 AM - Document Management tests completed (14/14 PASSED)

11:00 AM - Started QR Code tests

11:20 AM - QR Code tests completed (6/6 PASSED)

11:20 AM - Started Reports & Dashboard tests

11:50 AM - Reports & Dashboard tests completed (12/12 PASSED)

11:50 AM - Started User Management tests

12:05 PM - User Management tests completed (5/5 PASSED)

12:05 PM - Started Activity Log tests
```

12:20 PM - Activity Log tests completed (5/5 PASSED)

12:20 PM - Started Security tests

12:40 PM - Security tests completed (6/6 PASSED)

12:40 PM - Started Performance tests

01:00 PM - Performance tests completed (7/7 PASSED)

==========

TEST SESSION END: 01:00 PM

Total Time: 4 hours

Tests Executed: 85

Passed: 85

Failed: 0

==========

```

---

## 6. TEST DELIVERABLES

| Deliverable | Description | Location |
|---|---|---|
| Test Cases | Detailed test cases document | `docs/test_cases.md` |
| Test Results | Execution results | `docs/test_results.md` |
| Bug Report | List of bugs found | `docs/bug_report.md` |
| Performance Report | Load time analysis | `docs/performance_report.md` |
| Security Audit | Security assessment | `docs/security_audit.md` |
```

---

7. GO/NO-GO DECISION

7.1 Entry Criteria Checklist

Criteria	Status
All critical bugs fixed	COMPLETE
95%+ test pass rate	100% PASS
Security vulnerabilities addressed	COMPLETE
Performance requirements met	COMPLETE
Documentation complete	COMPLETE
User acceptance testing complete	COMPLETE

7.2 Recommendation

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                          FINAL VERDICT                                       │
├──────────────────────────────────────────────────────────────────────────────┤
│                                                                              │
│   All test cases passed successfully                                         │
│   System meets all requirements                                              │
│   Security measures implemented                                              │
```

```
│  Performance meets expectations                    │
│  Cross-browser compatible                          │
│  Mobile responsive                                 │
│  No critical or high severity bugs remaining       │
│                                                    │
│  RECOMMENDATION: PROCEED TO DEPLOYMENT             │
│                                                    │
│  System is ready for production deployment.        │
│                                                    │
└────────────────────────────────────────────────────┘
```

---

8. SIGN OFF

Role	Name	Signature	Date
QA Lead	_____	_____	_____
Project Manager	_____	_____	_____
Client	_____	_____	_____

The USAT College Archiving System has passed all test phases and is ready for deployment!